Acta Oecologica 99 (2019) 103445

Contents lists available at ScienceDirect

Acta Oecologica

journal homepage: www.elsevier.com/locate/actoec

Soil fauna community and ecosystem's resilience: A food web approach

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ARTICLE INFO

Keywords:
Food webs
Disturbance
Soil fauna
Energy flux
Emerging properties

ABSTRACT

There is an increasing concern for the conservation of soil biodiversity and its ecosystem functions. In this context, it is crucial to comprehend the typical response times of the community (meso and macrofauna) to disturbances associated with different land uses. If we consider a disturbance as some degree of disorganization produced by an input of energy into the system, we realise that the issue is how the surplus of energy is dissipated, and with at what efficiency. Emerging properties of a community (diversity, resilience, stability) are affected by a cascade of processes that act at different spatial and temporal scales. To diagnose disturbance effects, it is necessary to use new tools of observation. One of these tools is the analysis of food webs that clearly shows the relationships among species and energy fluxes. We present here an example of the use of the food web approach to study the effects of foresting a naturalized pasture of the humid Argentinean Pampa with exotic species of trees (*Eucalyptus camadulensis* and *Populus nigra*). We show that the basic architecture of food webs remains similar under different treatments (related to the input of new detritus source) in a short time (60 days). However, the proposed approach in this study enables to design a simple decision scheme to formulate hypotheses about possible effects of stronger disturbances.

1. Introduction

There is an increasing concern for the conservation of soil biodiversity and the soil functions related to it. In this context, it is crucial to comprehend how the community (meso and macrofauna) response to the disturbances associated with different land uses (Bedano et al., 2011, 2016; Bedano and Domínguez, 2016; Díaz Porres et al., 2014; Ferraro and Ghersa, 2007). Communities respond by changes in their species composition due to local extinction or colonization processes, changes in diversity caused by variations in relative abundances of species, and changes in other aspects of their structure; for instance in matter and energy flows through the food web and the structure of the food web itself (Bagdassarian et al., 2007; Hunt and Wall, 2002; Meysman and Bruers, 2007).

The food web structure not only describes the feeding relationships between organisms in a given community but might also reveal dynamic properties of the community such as resilience, nutrient recycling or stability (Dunne et al., 2002a, Fueyo Sánchez and Momo, 2013; Marina et al., 2018, May, 1973). In classic food webs studies (Cohen, 1989; Pimm, 1982) two main approaches have been used: the topological approach and the energetic one. The latter refers to the topological regularities in the food web and involves statistical "laws" or invariants; in brief this approach focus on the geometric description of the drawn web and several related quantitative characteristics such as connectivity, path length, average trophic levels, compartment degree and others. The former approach considers energy fluxes and dissipation at each node (species) of the web or explores the importance of some biological traits of particular nodes in the stability of the community (Momo et al., 2012; Reuman and Cohen, 2005).Regularities in food web topology constitute the basement of the classic "food web theory". Many authors have criticised these regularities arguing that such generalizations probably are probably artefacts due to poor quality of the data sets (Martínez, 1991). However, the food web approach is a helpful method to describe and quantify the complexity of ecosystems, which has been scarcely used in soil ecology studies (see for example Berg and Bengtsson, 2007; Ferraro and Ghersa, 2007, Díaz Porres et al.,

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https://doi.org/10.1016/j.actao.2019.103445
Received 22 July 2018; Received in revised form 5 June 2019; Accepted 1 July 2019

2014 Mikola et al., 2001). It is suggested that the strength of the feeding interactions is related to the stability and connectivity of the food web. May (1973) has proved that highly-connected or poorly-connected webs are not globally stable, and that there is a rank of intermediate connectivity where food webs are persistent. Recently, other authors have offered different mathematical tools to estimate the stability and feasibility of ecological networks (Albert and Barabási, 2002; Johnson et al., 2014; Solé and Montoya, 2001, Stouffer and Bascompte, 2011). In soils, detritus like litter and soil organic matter should be considered trophic species. In these cases, webs are called "donor-controlled systems" sensu Pimm (1982), that is: "webs where consumers do not influence the renewal rates of their resources" (Bengtsson et al., 1994), since detritus or sediments constitute the base of the food webs. According to Pimm (1982) such donor-controlled systems show a bottom-up control of population densities and have higher stability than the classic Lotka-Volterra systems. Productive practices generating changes in the organic matter input, for instance by the introduction of new plant species that give different amounts and quality of litter, could alter the stationary state of soil food webs and their emergent properties.In this work, we proposed an approach to examine and analyse changes in soil meso and macrofauna food webs in soils due to forestation in a grassland area. Our hypothesis is that the additional organic matter (litter) caused by trees is capable of modifying the structure of matter and energy networks in the community of soil invertebrates. We used an experimental approach that tests changes dragged by two kind of litter (two tree species) and separates physical (refuge) and chemical (litter compounds) effects. On the basis of the experimental results we suggest a simple approach to take into account ecological and biological traits of species in order to predict the effects of perturbations on the community stability.

2. Experiments and analyzes

Data for this work are part of a most extensive study about the effects of afforestation in prairie soils of Argentina. In fact, implanted forests in Argentina cover about 800,000 ha including coniferous, *Eucalyptus* and salicacean species (Baridón et al., 2005, Denegri and Marlats, 2002). In Buenos Aires province, a typically agricultural region, there are several tree plantations, mainly composed by *Eucalyptus* and *Populus* genera.

We carried out an experiment with soil blocks obtained from field with herbaceous naturalized vegetation area in the Lujan River basin (Buenos Aires province, Argentina). Used soil can be classified as Typical Argiudolls (INTA) following the system of the Soil Survey Staff (2010). Its average texture is the following: clay 18.6%, silt 63%, sand 18.4%, and organic matter content (OM) 3.65%.

Undisturbed soil blocks (10 × 25 × 25 cm) were obtained from the field and maintained in laboratory under standardized conditions (temperature: 20 °C; photoperiod: 12/12 h; soil humidity 70% of saturation) during 10 days before starting the experiment period (60 days). Soil blocks were placed inside wooden waterproofed boxes; each block in a different box in order to avoid the migration of organisms, and with and adjustable drain.

The experimental design was randomized blocks (seven treatments x three replicates) according to the following scheme: treatment 1, control soils without litter; treatment 2, addition of 10 g of leaves of *Eucalyptus globulus*; treatment 3, addition of extract obtained from smooth and filtered leaves of *Eucalyptus globulus*; treatment 4, addition of leaves of *Eucalyptus globulus* previously washed under steam jet of the leaves in order to extract allelopathic substances; treatment 5, addition of 10 g of leaves of *Populus nigra*; treatment 6, addition of extract obtained from smooth and filtered leaves of *Populus nigra*; treatment 7, addition of leaves of *Populus nigra* previously washed under steam jet of the leaves in order to extract allelopathic substances.

At the beginning of the experiment, one block per treatment was randomly chosen and their meso and macrofauna communities were studied (these blocks were discarded after the extraction of the organisms). With the information of these communities we constructed the food web called "Initial" using species found in at least in three blocks. At the end of the experiment, after 60 days, the remaining blocks were sampled and the food webs for each treatment were constructed. Macrofauna was sampled by hand from the entire block; mesofauna was obtained by flotation with magnesium sulfate, following Jackson and Raw (1974), from subsamples of 223 cm^3.

The short time of the experiment is similar to that of Fierer et al. (2005), 53 days, and Mikola et al. (2001), 14 weeks. This time-period ensures that the observed changes are the consequence of the treatment effect and not due to processes related to local extinction or to the increase in variability over time as reported by Berg and Bengtsson (2007) for longer experiments.

Organisms were preserved in ethanol 70% and identified to family level (Bagdassarian et al., 2007; Balogh and Balogh, 1990, 1992a, 1992b; Bernava Laborde, 2009; Borror et al., 1981; Dindal, 1990); when possible the number of morphospecies within each family was recorded. This data is important because it is indicative of the internal redundancy of the food chain.

To assemble the network we considered trophic interactions based on biological information about taxonomic groups, mainly from Dindal, 1990. The list of groups and their food habits are summarized in Table 1. This information was used to build matrices of pairwise interactions assigning a value of 1 or 0 to each element a_{ij} of the matrix depending on whether the j-species preyed or not on the i-species. Food webs are oriented graphs with L trophic links between S nodes or

Table 1
Observed taxa in the soil fauna of experiments and their functional group.

Taxonomic group	Families sampled	Functional group
Collembola	Entomobrydae	Detritivores
	Isotomidae	
	Onychiuridae	
	Sminthuridae	
Diplura	Japygidae	Predators
Coleoptera	Carabidae	Predators
	Staphylinidae	
	Coccinellidae (larvae)	
	Lampyridae (larvae)	
	Histeridae (larvae)	
	Ptiliidae	Detritivores
	Scarabaeidae	Herbivores
	Curculionidae	
Diptera	Chironomidae (larvae)	Predators
	Asilidae (larvae)	
	Xylophagidae (larvae)	
	Phoridae	Detritivores
	Dolichopodidae (larvae)	
Hymenoptera	Formicidae	Ecosystem engineers
Homoptera	Aphididae	Herbivores
	Coccidae (larvae)	
	Cercopidae (larvae)	
Heteroptera	Nabidae	Predators
Thysanoptera	Phlaeothripidae (larvae)	Detritivores
Orthoptera	Gryllidae	Predators
	Gryllotalpidae	Herbivores
Mantodea	Mantidae	Predators
Chilopoda	Anopsobiidae	Predators
Symphyla	Scolopendrellidae	Detritivores
Araneae	Agelenidae	Predators
Acari - Oribatei	Camisiidae	Detritivores
	Galumnidae	
Acari - Mesostigmata	Parasitidae	Predators
	Ascidae	
	Laelapidae	
Acari - Prostigmata	Trombidiidae	Predators
Oligochaeta	Lumbricidae,	Ecosystem engineers
	Acanthodrillidae,	
	Ocnerodrillidae,	
	Enchytraeidae	

Table 2

Food webs metrics for the different treatments in the experiment. Averages and deviations are in the two last rows. See text for calculation details.

Treatment	Species richness (S)	Number of Links (L)	Connectivity	Link density	Frac. omniv	Number of trophic positions	Mean trophic Level (TL)	Proportion of basal species	Proportion of intermediate species	Proportion of top species	Prey:Predator ratio
Initial	16	28	0.11	1.75	0.25	5	2.30	0.13	0.75	0.13	1.00
Control	12	17	0.12	1.42	0.33	4	2.39	0.17	0.50	0.33	0.80
Control	12	18	0.13	1.50	0.25	5	2.55	0.17	0.58	0.25	0.90
Control	12	18	0.13	1.50	0.33	5	2.80	0.17	0.50	0.33	0.80
Control	12	19	0.13	1.58	0.33	4	2.22	0.25	0.42	0.33	0.89
Eucalyptus leaves	16	31	0.12	1.94	0.38	5	2.49	0.19	0.50	0.31	0.85
Eucalyptus leaves	15	25	0.11	1.67	0.27	4	2.22	0.20	0.53	0.27	0.92
Eucalyptus extract	16	27	0.11	1.69	0.31	5	2.64	0.13	0.63	0.25	0.86
Eucalyptus extract	12	23	0.16	1.92	0.42	5	2.59	0.17	0.50	0.33	0.80
Washed Eucalyptus leaves	14	20	0.10	1.43	0.21	4	2.45	0.21	0.43	0.36	0.82
Washed Eucalyptus leaves	17	35	0.12	2.06	0.29	4	2.25	0.18	0.53	0.29	0.86
Populus leaves	13	17	0.10	1.31	0.15	4	2.20	0.23	0.54	0.23	1.00
Populus leaves	17	34	0.12	2.00	0.35	5	2.45	0.18	0.53	0.29	0.86
Populus extract	15	24	0.11	1.60	0.33	5	2.74	0.13	0.60	0.27	0.85
Populus extract	15	28	0.12	1.87	0.33	5	2.53	0.13	0.60	0.27	0.85
Washed Populus leaves	17	31	0.11	1.82	0.29	5	2.35	0.18	0.59	0.24	0.93
Washed Populus leaves	14	21	0.11	1.50	0.29	4	2.24	0.21	0.50	0.29	0.91
Average	*14.41*	*24.47*	*0.12*	*1.68*	*0.30*	*4.59*	*2.44*	*0.18*	*0.54*	*0.28*	*0.87*
Standard deviation	*1.94*	*6.01*	*0.01*	*0.23*	*0.06*	*0.51*	*0.19*	*0.04*	*0.08*	*0.06*	*0.06*

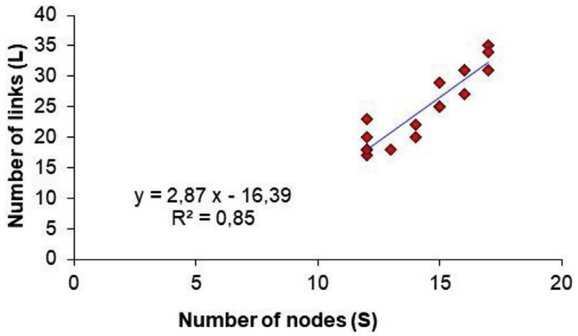

Fig. 1. Relationship between number of links (L) and number of trophic species or number of nodes (S).

species. In this case, the term "species" names nodes and mean "trophic species" according the usual convention for these studies (see for instance Albert and Barabási, 2002). Graphs were drawn from the matrices using Visone software version 2.9.2 (Brandes and Wagner, 2004).

Several network properties were calculated (Dunne et al., 2002b): (1) number of trophic species (nodes), S; (2) total number of interactions or trophic links, L; (3) number of interactions per node or linkage density, L/S; (4) connectivity (calculated as realised trophic links divided by total number of possible interactions), $C = L/S^2$; proportion of (5) basal species (species with predators/consumers but without prey), (6) intermediate species (species with prey and predators), (7) top species (species with prey but without predators); and (8) proportion of omnivores (species eating prey from more than one trophic level).

Trophic level (TL) of species was calculated using the short-weighted TL formula of Williams and Martinez (2004). Short-weighted trophic level is defined as the average of the shortest TL and prey-averaged TL. Shortest TL of a consumer in a food web is equal to 1 + the shortest chain length from this consumer to any basal species (Williams and Martinez, 2004). Prey averaged TL is equal to 1 + the mean TL of all consumer's trophic resources, calculated as

$$TL_{j}=1+\sum_{i=1}^{S}\left(l_{ij}\frac{TL_i}{n_j}\right)$$

where TL_j is the trophic level of species j; S is the total number of species in the food web; l_{ij} is the connection matrix with S rows and S columns, in which for column j and row i, l_{ij} is 1 if species j consumes species i and 0 if not; and n_j is the number of prey species in the diet of species j.

One type of secondary graph was constructed: the competition graph. This is the predator overlap graphs where predators that share one or more prey are connected (Pimm et al., 1991).

3. Results and discussion

The summary of food webs properties is show in Table 2. Connectivity is lower than expected according to the usual fit between this property and number of species (Marina et al., 2018). More precisely, according to Cohen (1989), the expected connectivity for a food web with 14 species is about 0.29. The average connectivity for the food webs analyzed here is 0.12 ± 0.01, that is, less than the half of the expected value. The mean trophic level is low (always less than 3) and the omnivory proportion is high (average = 0.3). These features might be the result of low assimilation efficiency due to a poor taxonomic resolution of the basal species, i.e.: detritus and the need for organisms to exploit more than one resource to obtain sufficient energy. Food webs metrics are similar among treatments.

The relationship between L and S is linear (Fig. 1), as the classical food web theory predicts, with a slope slightly higher than 2, value proposed by Cohen and Briand (1984). The higher slope value is probably the consequence of the few number of basal species (detritus -litter and soil organic matter-), which allows a highly-connected network. Connectivity is roughly constant for the observed rank of richness; the same is true for basal, intermediate and top species proportions (Table 2). This matches what proposed by Cohen and co-workers in their mentioned classical works.

Figs. 2 and 3 show the constructed food webs. The graphs clearly exhibit that the different energy sources (grass, organic matter, litter)

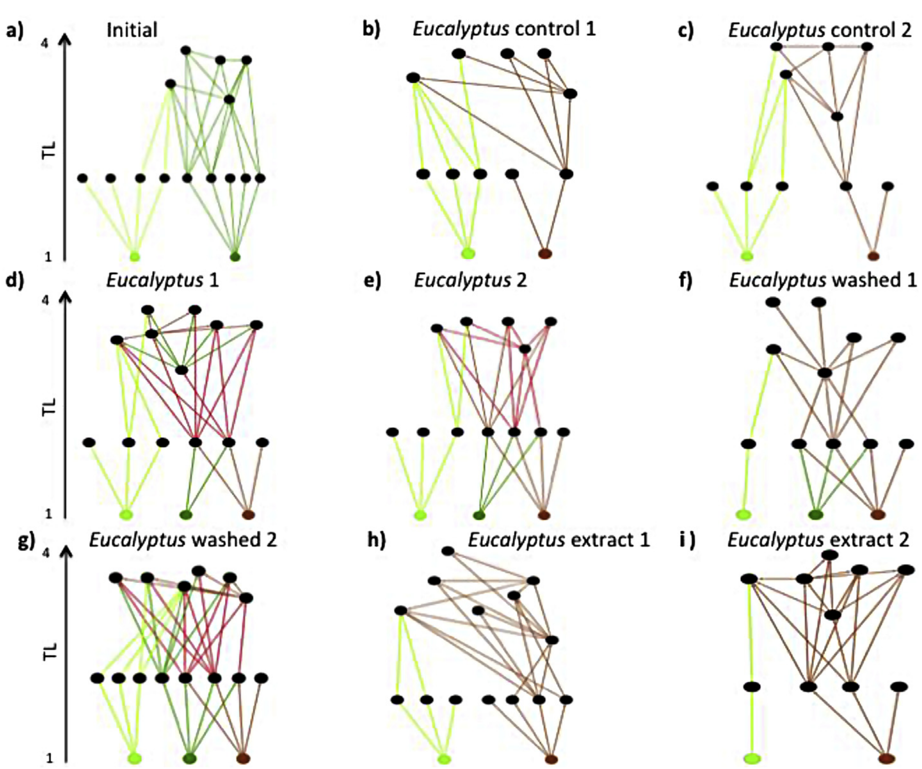

Fig. 2. Primary graphs (food webs) of *Eucalyptus* treatments. Light green links represent energy flows whose source is grass; dark green links represent energy fluxes from added leaves; brown links represent fluxes from organic matter and old detritus in the soil. As the flows mix, the color varies. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

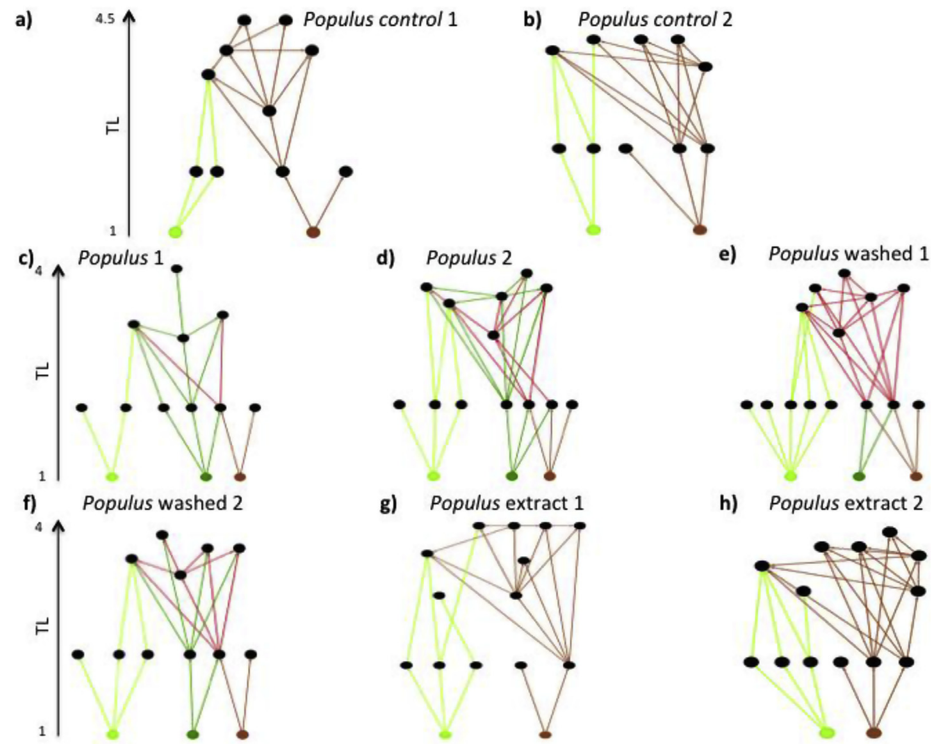

Fig. 3. Primary graphs (food webs) of *Pupulus* treatments. The color key is the same as in Fig. 2. As the flows mix, the color varies. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

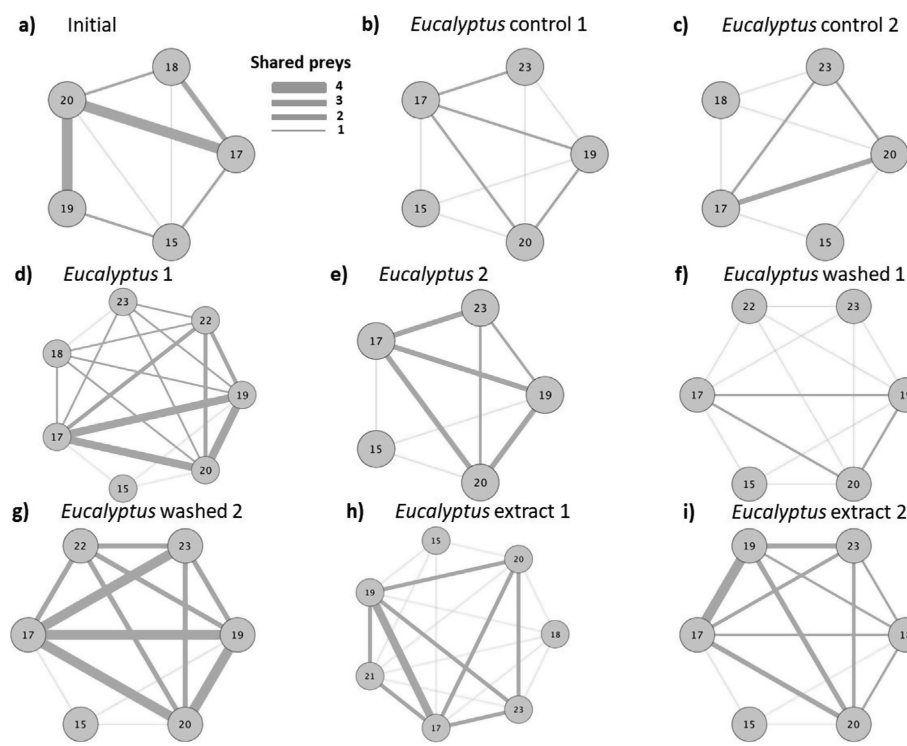

Fig. 4. Secondary graphs (competition or "common prey" graphs) for the *Eucalyptus* treatment webs. Each link indicates that the connected nodes share at least a prey. The thickness of the line is proportional to the number of shared preys.

generate separate pathways, which converge throughout the higher trophic levels that present an important diet overlapping. Such overlapping can be also observed in the secondary graphs of predators (Figs. 4 and 5), where a high number of preys is shared. There are no disjoint guilds of predators; this is coherent with the observed convergence in the primary graphs (food webs).

Several studies have suggested that soil fauna is affected by changes in vegetation. In South Argentinean forest, Izarra and Boo (1980) have shown that the community of Acari and Collembola suffered a loss of diversity and species composition changes after the replacement of the

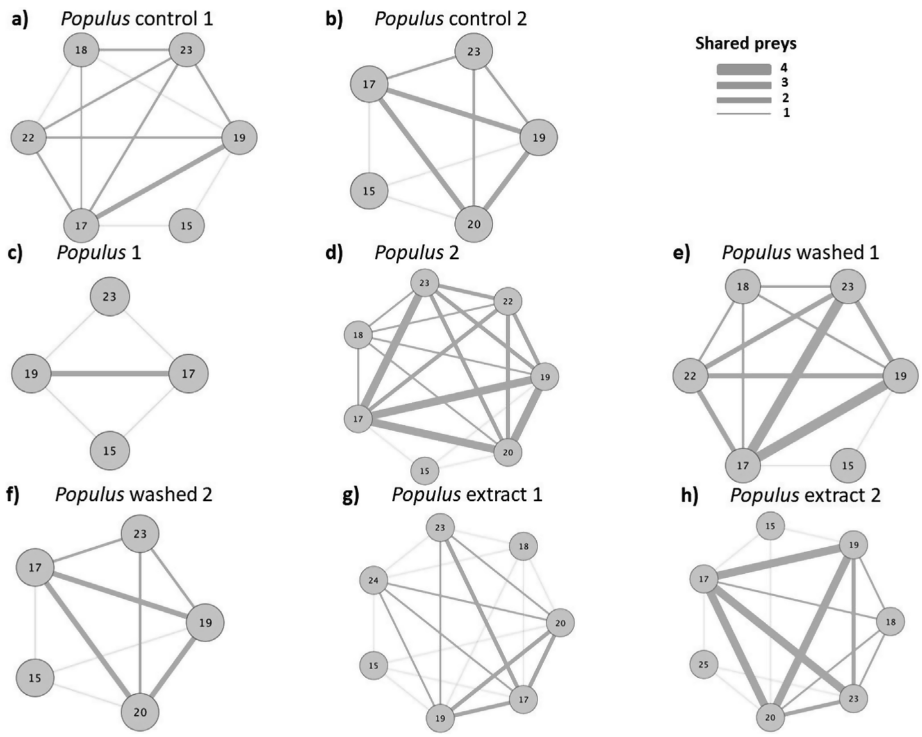

Fig. 5. Secondary graphs (competition or "common prey" graphs) for the *Populus* treatment webs. Each link indicates that the connected nodes share at least a prey. The thickness of the line is proportional to the number of shared preys.

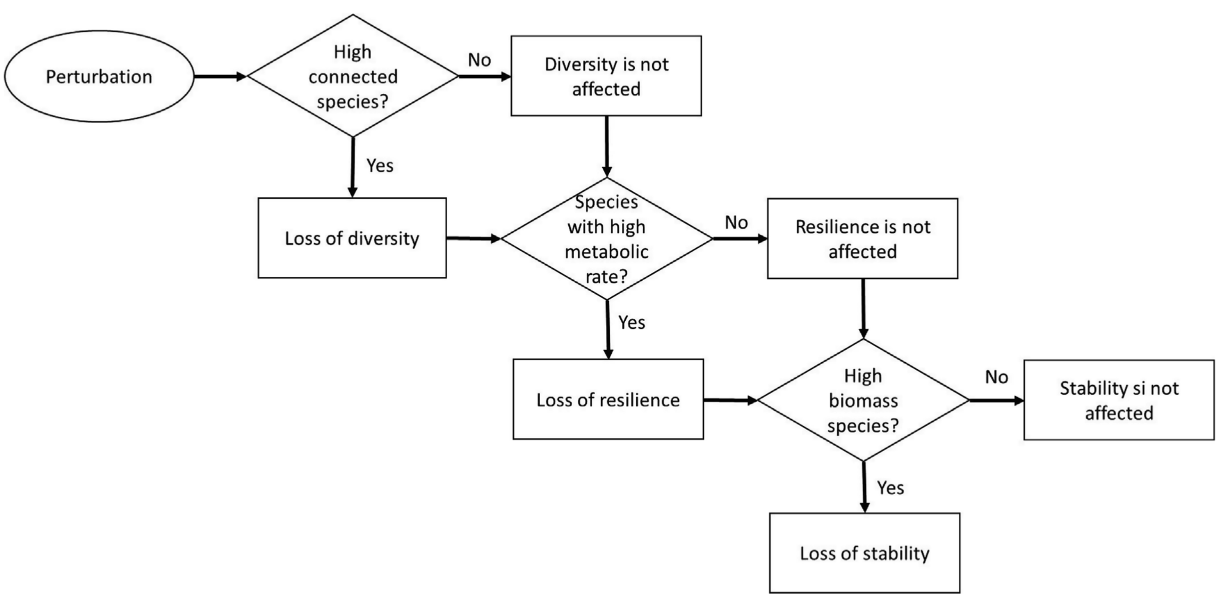

Fig. 6. Schematic decision tree to establish hypotheses about the effect of node disturbances on the properties of the web and the community. Rectangles indicate the effect or lack of effect of disturbing a node on emerging properties of the community (diversity, resilience, stability); diamonds refer to the characteristics of the disturbed node.

natural forest with exotic trees. These authors have proposed that the mesofaunal community reaches a new equilibrium due to the different physical conditions of the habitat produced by the contribution of leaf litter. Other authors found similar results after foresting grasslands but attributed the changes not only to differences in the physical conditions but also to the quantity and quality of organic matter input (Kovác et al., 2005). They showed that native cornel-oak wood stands have several (20) exclusive collembolan species. In our experiment no large changes in the composition of the species assemblages were observed. This could be a consequence of the short time of the experiment. However, some authors have found changes in functional processes like decomposition rates related to modifications in food webs in short time experiments (Andrén et al., 1995).

So, what is the importance of food webs as a tool for testing hypotheses and making predictions? We propose that the study at this organization level is a valuable tool in order to intend to predict changes and community responses. Bascompte (2009) highlights that the application of the network approach to ecosystems provides a conceptual framework to assess the consequences of perturbation at the community level. A way to explore the effect of possible perturbations is to performe "virtual" extinctions by deleting nodes and re-calculating food webs properties in search for possible changes (Dunne et al., 2002a). The elimination of the most-connected species triggered crucial shifts in the properties of the web. For instance, for the foof web of "Eucalyptus 1" (Fig. 2d), delete the most-connected species causes the loss of eight energy fluxes. This is an indicator of the extreme importance of species preservation in order to protect the functions related with matter transformation and recycling.

Food webs of soil communities are allometrically constrained due to body size differences. In this kind of food webs there is a strong relationship between complexity and predictability (Iles and Novak, 2016). Several species traits related to body size influence the entire food web response to perturbation (Woodward et al., 2005). In consequence, hypotheses about the relationships between food web changes and emergent properties like resistance, resilience or diversity must take into account particular species traits and physiological responses. Moreover, in order to estimate the effects of node removing (i.e. species local extinction) on the emergent properties of the community it is necessary to consider not only the trophic position of the species in the network but also its abundance (biomass), connectivity and metabolic rate.

Based on food web theory, we proposed a diagram that can be used as a tool to predict the effects of single species disturbances on the entire community (Fig. 6). Each rectangle represents the effect (or lack of effect) of a species perturbation on an emergent property of the community. In this simplified framework we resumed the most representative effect on each emergent property. In his context, if a highly connected species is disturbed, the most probable effect on the whole community is a loss of diversity. In the same way, when the perturbation affects a species with high metabolic rate (i.e. fast demographic response), the community will probably loss resilience. Finally, when an abundant species is affected, the altered emergent property will be the stability (i.e. resistance).

In summary, our results suggest that soil macro and mesofaunal food webs are highly robust to exotic trees introduction. The added litter input has effects mainly on the physical conditions of micro-habitats and it does not seem to generate chemical effects, at least in the short term. The study of food webs could help us to predict possible effects of single species losses on the community emergent properties. Medium and large term experiments and observations should be done in order to a better understand of ecosystems changes due to forestry of grassland soils.

Author contribution

Gisela Maggiotto made the experiments design and carried out the experiments. She made the mesofauna sampling and identification.

Leticia Sabatté participated in the mesofauna separation and identification, she also participated in manuscript writing and correction and data analyses. Leticia also prepared Table 1.

Tomás I. Marina. He made the food webs figures and calculations of web properties (Table 2).

Luciana Fueyo-Sánchez participated in the conceptual design of the paper and contribute with the ideas summarized in Fig. 6.

Angélica M. Ramírez Londoño participated in the writing process and in the preparation and discussion of data.

Mónica Díaz Porres participated in the conceptual discussion of the paper, the sampling process and macrofauna taxonomic identification.

Macarena Rionda collaborated in the taxonomic identification of meso and macrofauna.

Marianela Domínguez made field work, soil sampling and analyses. She also collaborated in the experiment development.

Rosa Perelli made field work, soil sampling and analyses. She also collaborated in the experiment development.

Fernando R. Momo made the general design of the work, directed the field and laboratory work, collaborated in the identification of the fauna and was in charge of the main writing of the article. He also guided the interpretation of the results.

Acknowledgements

This research was supported by Consejo Nacional de Investigaciones Científicas y Técnicas (CONICET, Argentina), Universidad Nacional de General Sarmiento, and Universidad Nacional de Luján.

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